



ArrayList Java Cheat Sheet

◇ 1. Import & Create

```
import java.util.ArrayList;
```

```
ArrayList<Integer> list = new ArrayList<>();  
ArrayList<String> names = new ArrayList<>();
```

👉 Dynamic array (resizable, unlike normal arrays)

◇ 2. Add Elements

```
list.add(10);           // add at end  
list.add(1, 20);        // add at index
```

◇ 3. Access Elements

```
int val = list.get(0);  // get element
```

◇ 4. Update Elements

```
list.set(0, 99);        // replace index 0 with 99
```

◇ 5. Remove Elements

```
list.remove(0);          // remove by index  
list.remove(Integer.valueOf(10)); // remove by value
```

◇ 6. Size & Check

```
list.size();           // number of elements
list.isEmpty();        // true/false
list.contains(10);     // check element
```

👉 contains() checks if element exists

◇ 7. Loop / Traverse

```
// for loop
for (int i = 0; i < list.size(); i++) {
    System.out.println(list.get(i));
}
```

```
// for-each
for (int x : list) {
    System.out.println(x);
}
```

◇ 8. Sorting

```
import java.util.Collections;
```

```
Collections.sort(list);           // ascending
Collections.sort(list, Collections.reverseOrder()); // descending
```

◇ 9. Other Useful Methods

```
list.clear();           // remove all
list.indexOf(10);        // first index
list.lastIndexOf(10);    // last index
```

```
list.toArray();           // convert to array
list.addAll(otherList);    // merge lists
```

◇ 10. Convert Array ↔ ArrayList

```
// Array → ArrayList
String[] arr = {"A", "B"};
ArrayList<String> list = new ArrayList<>(Arrays.asList(arr));
```

```
// ArrayList → Array
String[] newArr = list.toArray(new String[0]);
```

◇ 11. Key Properties (💧 important for exams)

- Dynamic size (grow/shrink)
- Allows duplicates
- Maintains order
- Fast access ($O(1)$)
- Slow insert/delete in middle ($O(n)$)

🧠 Pro Tip (interview gold)

```
List<Integer> list = new ArrayList<>();
```

👉 Always prefer **List interface** instead of direct ArrayList (flexibility magic ✨)

⚡ Mini Example

```
ArrayList<Integer> list = new ArrayList<>();

list.add(1);
list.add(2);
```

```
list.add(3);
```

```
list.remove(1);
```

```
for (int x : list) {  
    System.out.println(x);  
}
```